

Lecture 9: Neutron Interactions and Cross Sections

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

Spring 2026

1 Introduction: The Shift to Induced Reactions

Primary Reference: Lamarsh, 4th Ed., Section 3.1

In the first phase of this course, we studied *spontaneous* nuclear transformations—radioactive decay ($dN/dt = -\lambda N$). We are now shifting to *induced* reactions, where we actively bombard nuclei with projectiles to force a transformation.

While alpha particles and protons were used in early experiments, they suffer from a major limitation: the **Coulomb Barrier**. To enter a nucleus, a charged particle must possess enough kinetic energy to overcome the electrostatic repulsion of the target protons.

Neutrons, being neutral, face no such barrier. A neutron with negligible kinetic energy ($E \sim 0.025$ eV) can wander into a uranium nucleus as easily as a fast neutron ($E \sim 2$ MeV). This property makes the neutron the ideal agent for nuclear engineering.

2 The Microscopic Cross Section (σ)

Primary Reference: Lamarsh Section 3.2

When a neutron approaches a nucleus, the probability of an interaction depends on the effective "target area" presented by the nucleus. We call this the **Microscopic Cross Section**, denoted by σ .

2.1 The Barn

The standard unit for cross sections is the **barn** (b).

$$1 \text{ b} = 10^{-24} \text{ cm}^2 \quad (1)$$

This value is roughly the geometric cross-sectional area of a medium-sized nucleus. However, the *interaction* cross section is a quantum mechanical property and can be vastly different from the geometric size.

Yes, the unit name really did come from midwest hunting prowess (or lack thereof). An amusing story of how it came to be named is given [here](#).

3 Classification of Interactions

Reference: Lamarsh Section 3.1 - 3.2

A neutron does not just "hit" a nucleus; it interacts via specific channels. The **Total Cross Section** (σ_t) is the sum of the probabilities of all possible distinct events.

$$\sigma_t = \sigma_s + \sigma_a \quad (2)$$

3.1 1. Scattering (σ_s)

The neutron interacts with the nucleus and re-emerges, potentially with lower energy.

- **Elastic Scattering (σ_e or (n, n)):** "Billiard ball" collision. Kinetic energy is conserved in the center-of-mass system. This is the primary mechanism for slowing down neutrons in a moderator (e.g., Water, Graphite).
- **Inelastic Scattering (σ_i or (n, n')):** The neutron strikes the nucleus and leaves it in an excited state. The neutron loses significant kinetic energy. This generally requires high-energy neutrons (Fast).

3.2 2. Absorption (σ_a)

The neutron enters the nucleus and does not re-emerge as a neutron.

- **Radiative Capture (σ_γ or (n, γ)):** The neutron is captured, and the nucleus emits a gamma ray. (Example: $^{238}\text{U} + n \rightarrow ^{239}\text{U} + \gamma$).
- **Fission (σ_f or (n, f)):** The nucleus splits into two large fragments plus 2–3 neutrons.
- **Charged Particle Emission (σ_p, σ_α):** Example: $^{10}\text{B} + n \rightarrow ^7\text{Li} + \alpha$ (Control Rods).

4 The Macroscopic Cross Section (Σ)

Primary Reference: Lamarsh Section 3.3

In engineering, we deal with bulk matter. We define the **Macroscopic Cross Section** (Σ) to describe the interaction probability per unit volume.

$$\Sigma = N\sigma \quad (3)$$

where N is the atom density (atoms/cm³). The unit of Σ is **cm⁻¹**, representing the probability of interaction per cm of travel.

4.1 Calculating Atom Density (N)

For a pure substance with density ρ (g/cm³) and atomic weight M (g/mol):

$$N = \frac{\rho N_A}{M} \quad (4)$$

For a mixture (like H₂O), the macroscopic cross section is additive:

$$\Sigma_{mix} = \sum_i N_i \sigma_i \quad (5)$$

Example 1: Macroscopic Cross Section of Water

Problem: Calculate the total macroscopic cross section (Σ_t) for liquid water at 20°C. Assume: $\rho = 1.0 \text{ g/cm}^3$; $\sigma_t^H = 21.0 \text{ b}$; $\sigma_t^O = 4.0 \text{ b}$.

Solution: 1. Calculate Molecular Density of Water (N_{H_2O}):

$$N_{H_2O} = \frac{\rho N_A}{M} = \frac{(1.0)(0.6022 \times 10^{24})}{18.0} = 0.0335 \times 10^{24} \text{ molecules/cm}^3$$

2. Determine Atom Densities (N_H and N_O):

$$N_O = 1 \times N_{H_2O} = 0.0335 \times 10^{24} \text{ atoms/cm}^3$$

$$N_H = 2 \times N_{H_2O} = 0.0670 \times 10^{24} \text{ atoms/cm}^3$$

3. Calculate Σ_t :

$$\Sigma_t = N_H \sigma_t^H + N_O \sigma_t^O$$

$$\Sigma_t = (0.0670)(21.0) + (0.0335)(4.0) \quad [\text{units of } 10^{24} \cdot 10^{-24} \text{ cancel}]$$

$$\Sigma_t = 1.407 + 0.134 = \mathbf{1.54 \text{ cm}^{-1}}$$

Note: Hydrogen contributes over 90% of the interaction probability. The collision cross section listed is for free hydrogen and appropriate for higher energy neutrons. For low energy neutrons the collision cross section for hydrogen bound up in water is five times greater.

5 Beam Attenuation and Mean Free Path

Consider a collimated beam of neutrons with intensity I_0 incident on a slab. The removal of neutrons follows Beer's Law:

$$I(x) = I_0 e^{-\Sigma_t x} \quad (6)$$

The average distance a neutron travels before interacting is the **Mean Free Path** (λ):

$$\lambda = \frac{1}{\Sigma_t} \quad (7)$$

Example 2: Shielding Calculation

Problem: An experiment requires a neutron beam to be shielded by a slab of Indium ($\Sigma_a = 4.8 \text{ cm}^{-1}$). How thick must the slab be to reduce the beam intensity by 95%?

Solution: If intensity is reduced *by* 95%, the remaining intensity $I(x)$ is **5%** of I_0 .

$$\frac{I(x)}{I_0} = 0.05$$

$$0.05 = e^{-\Sigma x} \implies \ln(0.05) = -\Sigma x$$

$$-2.996 = -4.8x \implies x = \frac{2.996}{4.8} = \mathbf{0.624 \text{ cm}}$$

6 Reaction Rates and Flux

Primary Reference: Lamarsh Section 3.4

In a reactor, neutrons move randomly in all directions. We define the **Scalar Flux**, $\phi = nv$, where n is neutron density and v is speed. The **Reaction Rate Density** (R) is the fundamental link between physics and engineering:

$$R = \phi \Sigma \quad (8)$$

(Units: interactions $\cdot \text{cm}^{-3} \cdot \text{s}^{-1}$)

Example 3: Measuring Flux with Gold Foil

Problem: A gold foil (mass 1.0 mg) is irradiated in a reactor. The foil becomes radioactive at a rate of 10^8 interactions/sec (R_{total}). Calculate the neutron flux ϕ . Assume: $\rho_{Au} = 19.3 \text{ g/cm}^3$; $M_{Au} = 197 \text{ g/mol}$; $\sigma_\gamma = 98.7 \text{ b}$.

Solution: The total reaction rate in the foil is $R_{total} = \phi(\text{Total Atoms})\sigma_\gamma$. 1. Calculate Total Atoms:

$$\text{Atoms} = \frac{mN_A}{M} = \frac{(0.001)(0.6022 \times 10^{24})}{197} = 3.057 \times 10^{18} \text{ atoms}$$

2. Solve for ϕ :

$$10^8 = \phi \cdot (3.057 \times 10^{18}) \cdot (98.7 \times 10^{-24})$$

$$10^8 = \phi \cdot (3.017 \times 10^{-4})$$

$$\phi = \mathbf{3.31 \times 10^{11} \text{ neutrons/cm}^2\text{s}}$$

7 Application of Cross-Sections: Boron Neutron Capture Therapy (BNCT)

An elegant application of neutron absorption is found in cancer treatment, specifically **Boron Neutron Capture Therapy (BNCT)**. This technique relies on the balance between atomic density (N) and absorption cross-sections (σ).

7.1 The Primary Mechanism

1. A boron-containing compound (e.g., BPA [Boronophenylalanine: metabolic targeting] or BSH [Sodium Borocaptate: leaky BBB targeting]) is administered to the patient. These compounds are chemically designed to accumulate preferentially in or adjacent to tumor cells.
2. The tumor is irradiated with neutrons.
3. **The Kill Reaction:** Boron-10 captures a thermal neutron and immediately splits into two heavy, charged particles:



(Note: γ emitted in only 6% of reactions, reducing recoil KE in those cases)

7.2 The Competition: Microscopic vs. Macroscopic

The physical challenge is that the human body and tumor consists mostly of Hydrogen, Carbon, Nitrogen, and Oxygen, with only trace amounts of Boron (typically ~ 30 ppm). The probability of interaction is determined by the **Macroscopic Cross-Section** $\Sigma = N\sigma$.

- **Tissue (H, N):** Extremely high atomic density (N), but low microscopic cross-section ($\sigma \approx 0.3 - 1.8$ b).
- **Boron-10:** Extremely low atomic density (N), but massive microscopic cross-section ($\sigma \approx 3,840$ b).

Because σ_{Boron} is orders of magnitude larger than σ_{Tissue} , the Boron is able to compete with the bulk tissue for neutron capture, even at low concentrations. This creates a favorable **Therapeutic Ratio**, where the lethal high-LET dose (from the α and Li particles) is delivered primarily to the tumor cells, while healthy tissue receives a lower background dose.

7.3 Parasitic Absorption (Background Dose)

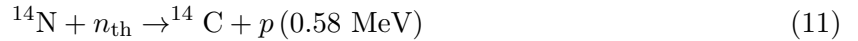
Despite the high cross-section of Boron, the prevalence of Hydrogen and Nitrogen means "side reactions" are unavoidable. These set the limit on the maximum radiation dose a patient can tolerate:

1. The Gamma Generator (Hydrogen Capture):



Hydrogen has a small σ (0.33 b), but it is the most abundant element in the body. This capture releases a penetrating gamma ray that irradiates healthy tissue far from the interaction site.

2. The Proton Generator (Nitrogen Capture):



Nitrogen contributes a significant background dose because the ejected proton creates localized damage similar to the Boron alpha particle, but indiscriminately in both healthy and tumor tissue. Because of the higher LET of the high energy proton emitted by Nitrogen than the γ emitted by Hydrogen, this is often the dominant non-Boron dose.

7.4 The Engineering Limit: The T/N Ratio

Ideally, Boron would accumulate *only* in the tumor. In reality, chemical kinetics limit us to a differential concentration, defined as the **Tumor-to-Normal (T/N) Ratio**.

- Typical clinical T/N Ratio: $\approx 3 : 1$.
- **The Consequence:** Healthy tissue also contains Boron (approx. 10 ppm).

The total dose delivered to the patient is a sum of the "Background Dose" (from Hydrogen and Nitrogen capture in healthy tissue) and the "Boron Dose" (from Boron capture in healthy tissue). The treatment must be terminated when the healthy tissue reaches its radiation limit, hopefully after the tumor (which received $3\times$ the Boron dose) has been destroyed.

7.5 The Physics Challenge: Moderation

A major engineering hurdle is delivering thermal neutrons to deep-seated tumors (like glioblastomas).

- **Thermal Neutrons (0.025 eV):** Have a high probability of absorbing in Hydrogen within the first few centimeters of tissue. (Historical trials often required removing the skull to irradiate the brain directly [3]).
- **Epithermal Neutrons (1 eV – 10 keV):** Modern sources use higher energy beams that penetrate the skull and skin. As they travel through tissue, they collide with Hydrogen and moderate (slow down), becoming thermal neutrons at the depth of the tumor.

References

- [1] Wikipedia contributors. "Neutron capture therapy of cancer." *Wikipedia, The Free Encyclopedia*.
- [2] Barth, R. F., Coderre, J. A., Vicente, M. G. H., & Blue, T. E. (2005). [Boron neutron capture therapy of cancer: current status and future prospects](#). *Clinical Cancer Research*, 11(11), 3987-4002.
- [3] Sweet, W. H., & Javid, M. (1952). [The possible use of neutron-capturing isotopes such as boron-10 in the treatment of neoplasms](#). *Journal of Neurosurgery*, 9(2), 200-209.

8 Resources for Visualization

1. **NNDC Sigma Plotter:** To see the difference between Capture (σ_γ) and Fission (σ_f) cross sections for U-235:
<https://www.nndc.bnl.gov/sigma/> (Select "n,g" for capture and "n,f" for fission).
2. **MIT OpenCourseWare (22.01):** Lecture 21 covers Neutron Interactions.
<https://ocw.mit.edu/courses/22-01-introduction-to-nuclear-engineering-and-ionizing-radiation/pages/syllabus/>
3. **HyperPhysics:** Cross Section Concepts.
<http://hyperphysics.phy-astr.gsu.edu/hbase/Nuclear/crosec.html>

How to use the NNDC Sigma Plotter

The National Nuclear Data Center (NNDC) provides the standard evaluated cross-section libraries (ENDF/B) used by nuclear engineers.

1. **Access the Tool:** Go to <https://www.nndc.bnl.gov/sigma/>.
2. **Method A: Visual Search (The Periodic Table)**
 - Click on the element symbol in the Periodic Table (e.g., U for Uranium).
 - A list of isotopes will appear. Click the specific isotope mass number (e.g., **235**).
 - A table of "Reactions" will appear.

3. Method B: Direct Search (Faster)

- Select the "Basic Retrieval" tab.
 - In the search bar ("Target"), simply type the isotope name (e.g., U235 or Fe56).
 - **Selecting Reactions:** Select the checkboxes for the cross sections you wish to compare:
 - **(n,tot):** Total Cross Section (σ_t)
 - **(n,el):** Elastic Scattering (σ_s)
 - **(n,g):** Radiative Capture (σ_γ)
 - **(n,f):** Fission (σ_f)
 - Click on "Submit"
4. **Plotting:** Click the **Plot** button. Use the axis settings to switch to Log-Log scales for better visibility of resonances.

Alternatives to NNDC

The standard NNDC site is often offline. Another site is from the Korea Atomic Energy Research Institute.

Primary Tool: KAERI Table of Nuclides <https://atom.kaeri.re.kr/nuchart/>

Exercise for Students:

1. Go to the URL above.
2. In the "Nuclide" search box, type: U238 (and hit Enter).
3. **Choose the Library:** Locate the column labeled **ENDF/B-VIII.0** (this is the US standard) and click on the small + sign to expand the options.
4. **Select the Data:** In the expanded list, locate the row labeled **Capture cross sections** and click the button labeled **Plot**.
5. **Verify Scale:** The graph that appears is the (n, γ) radiative capture cross section. The axes should default to **Log-Log** scale. (If they appear linear, use the settings to switch them to Log).
6. **Observation:** Look at the energy range between 10^0 eV and 10^4 eV. You will see the massive peaks rising 1000x above the baseline. These are the "Resonance Traps" our moderator design must defeat.

Recommended Exercise: The "Iron Window" (Bug or Feature?)

To understand why material properties can be both a hazard and a tool, try this:

- **Search:** Go to the IAEA or NNDC plotter and search for **Fe-56** (the main isotope in steel).
- **Plot:** Select the Total Cross Section **(n,tot)**.
- **Observation:** Zoom in on the region around **24 keV**. You will see a deep "valley" where the cross section drops precipitously (from ~ 10 b down to ~ 1 b).

- **The Shielding Nightmare (The Bug):** This is known as an interference minimum. If you are building a shield for a reactor, this "window" allows 24 keV neutrons to penetrate steel far more effectively than neighboring energies.
- **The Experimental Opportunity (The Feature):** Conversely, if you placed a 30 cm block of iron in front of a reactor beam, what would come out the other side? *A narrow-band beam strongly peaked near 24 keV.* Physicists use this "Iron Filter" technique to create quasi-monoenergetic beams for calibration.